Routing design -- OSPF underlay per site, eBGP overlay over two IPsec transports

Internet transports

ISP-A AS 65100
default-originate only

ISP-B AS 65200
default-originate only

eBGP

eBGP

Tu13 10.0.0.16/30
eBGP v4+v6

Tu23 10.0.0.20/30
eBGP v4+v6

AS 65020 -- HAN
announces 10.20.0.0/16 + 2001:db8:a020::/48 only

iBGP (Lo0) + OSPF

HAN-EDG1
10.20.255.1

HAN-EDG2
10.20.255.2

OSPFv2/v3 process 20, area 0
Lo0s, interlink, LAN sub-interfaces (passive)

VLAN 20/40/60/99

Tu11 10.0.0.0/30
eBGP v4+v6

Tu21 10.0.0.8/30
eBGP v4+v6

Tu12 10.0.0.4/30
eBGP v4+v6

Tu22 10.0.0.12/30
eBGP v4+v6

AS 65030 -- DAD
announces 10.30.0.0/16 + 2001:db8:a030::/48 only

iBGP (Lo0) + OSPF

DAD-EDG1
10.30.255.1

DAD-EDG2
10.30.255.2

OSPFv2/v3 process 30, area 0
Lo0s, interlink, LAN sub-interfaces (passive)

VLAN 30/40/50/60/99

AS 65010 -- HCM
announces 10.10.0.0/16 + 2001:db8:a010::/48 only

HCM-EDG1
10.10.255.1

iBGP (Lo0) + OSPF

HCM-EDG2
10.10.255.2

OSPFv2/v3 process 10, area 0
E2 default + remote /16s redistributed from eBGP at the edges

HCM-CORE1
10.10.255.3

HCM-CORE2
10.10.255.4

Plane A (blue): EDG1s over ISP-A
Plane B (orange): EDG2s over ISP-B
GRE + IPsec (IKEv2, AES-256, SHA-256)
BGP timers 5/15 s per tunnel
Inbound: enterprise /16s only
Outbound: own site aggregate only
(no transit, no default via tunnels)

Failover paths
1. Tunnel/ISP-A down -> plane B
2. EDG1 down -> HSRP + iBGP to EDG2
3. Core link down -> OSPF ECMP
Internet: local PAT at every edge
ISP sends default only